

Introduction to **5 BOARDS** you must know

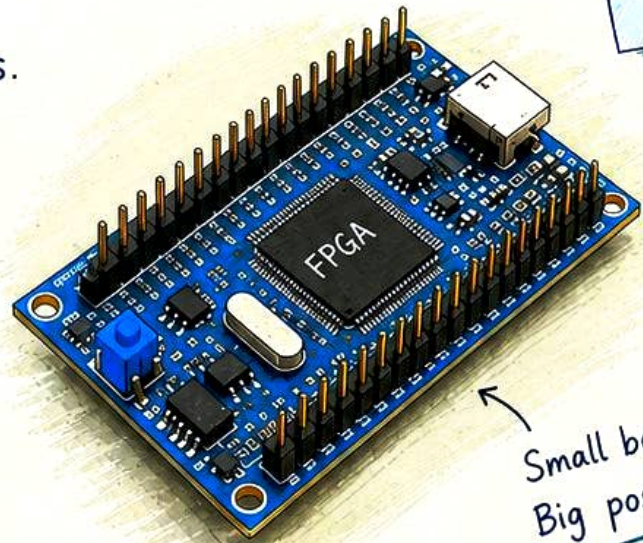
In this series, we will explore
5 powerful development boards
that are widely used in
embedded systems and IoT projects.

Each board has
its own strengths
and unique features...

Let's
explore
them!
😊

Why these boards?

- ✓ High performance
- ✓ Low power consumption
- ✓ Rich peripherals
- ✓ Real-time capabilities
- ✓ Wide ecosystem & support



Small boards...
Big possibilities!

The 5 Boards

STM32
(Blue Pill)

Arduino
Uno

ESP32
Dev Board

Raspberry Pi
Pico

FPGA

“Different boards, same goal —
to turn your ideas into real projects.”



Small board...
Big possibilities!

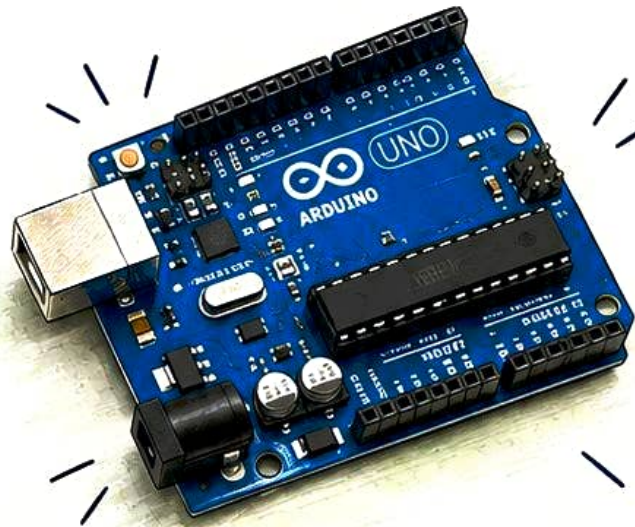
Arduino UNO

Open Source • Simple • Powerful

💡
Turn your
ideas into
reality!

What is Arduino Uno?

- Arduino Uno is a microcontroller board based on the **ATmega328P**.
- It is an open-source platform used for building and testing electronic projects.
- It is beginner-friendly and widely used by hobbyists, students and professionals.



The
heart of
your project

⚙️ Key Features

- ATmega328P microcontroller
- 14 digital I/O pins (6 PWM)
- 6 analog input pins
- 16 MHz crystal oscillator
- USB interface (for programming)
- Power via USB or external supply
- Simple and easy to use

🔌 Pinout (Main)

- Digital Pins : 0 - 13 (6 PWM)
- Analog Pins : A0 - A5
- Power Pins : 5V, 3.3V, GND, VIN, RESET
- Communication : TX (1), RX (0)

🎯 Applications

- Robotics & automation
- Sensor data logging
- Home automation
- IoT projects
- Educational purposes

Easy to
learn,
fun to
build!

⚡ Power Supply

- USB (5V)
- External (7 - 12V)

Same board...
endless projects! 😊



STM32

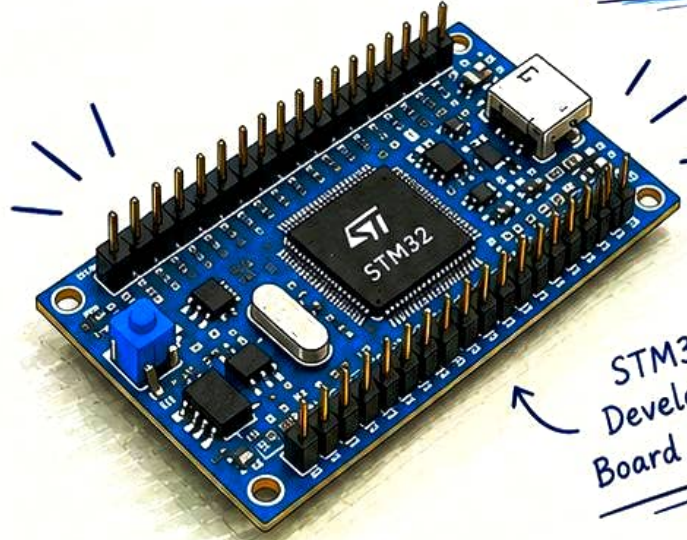
Powerful • Flexible • Reliable

Small chip...
Big possibilities!
😊



What is STM32?

- STM32 is a family of 32-bit microcontrollers based on ARM Cortex-M cores, known for high performance and low power.



STM32 Development Board (Blue Pill)



Key Features

- High performance ARM Cortex-M core
- Low power consumption
- Rich peripheral set
- Real-time capabilities
- Wide ecosystem & support



Why Choose STM32?

- Good balance of performance and power.
- Large community support.
- Easy to program and develop.
- Suitable for both beginners and professionals.



Applications

- Industrial automation
- Home automation
- IoT devices
- Robotics
- Consumer electronics

Build smarter projects!



Pinout (Main)

- Digital Pins : 0 – 53 (GPIO)
- Analog Pins : A0 – A7 (ADC)
- Power Pins : 3.3V, GND, VIN, RESET



“Learn • Build • Create”





ESP32

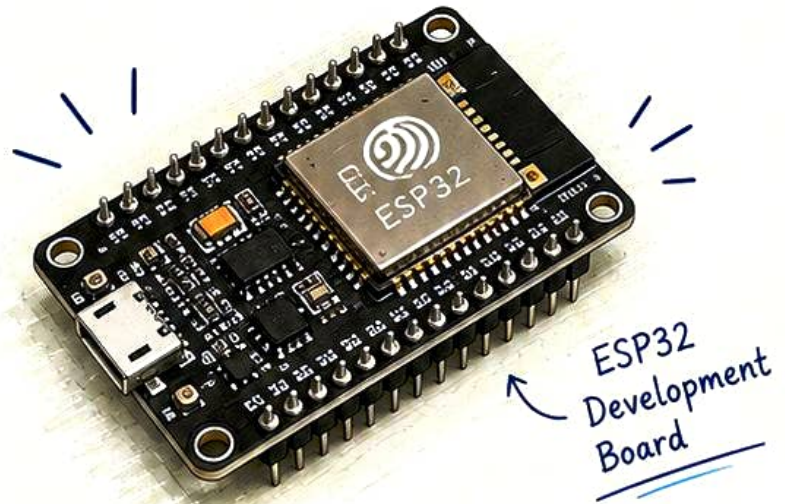
Versatile • Powerful • Connected

Small chip...
Big possibilities!
😊



What is ESP32?

- ESP32 is a powerful dual-core microcontroller developed by Espressif Systems. It comes with built-in Wi-Fi and Bluetooth, making it ideal for IoT and wireless projects.



Key Features

- Dual-core processor (up to 240 MHz)
- Built-in Wi-Fi (802.11 b/g/n)
- Bluetooth (Classic & BLE)
- Low power consumption
- Rich peripheral set
- Supports multiple programming environments (Arduino, ESP-IDF, MicroPython)



Why Choose ESP32?

- Integrated Wi-Fi & Bluetooth
- Excellent performance at low cost
- Great for IoT and smart devices
- Wide community support
- Flexible and easy to use



Applications

- IoT projects (Home automation, smart devices)
- Wearables
- Robotics
- Wireless sensor networks
- Smart agriculture & healthcare

Build smarter with ESP32!



Pinout (Main)

- Digital GPIOs : 0 – 39
- Analog (ADC) : 0 – 7 (12-bit)
- DAC : 2 (GPIO25, 26)
- Power : 3.3V, GND, VIN

“Learn • Build • Create”

Raspberry Pi Pico

Small • Powerful • Affordable

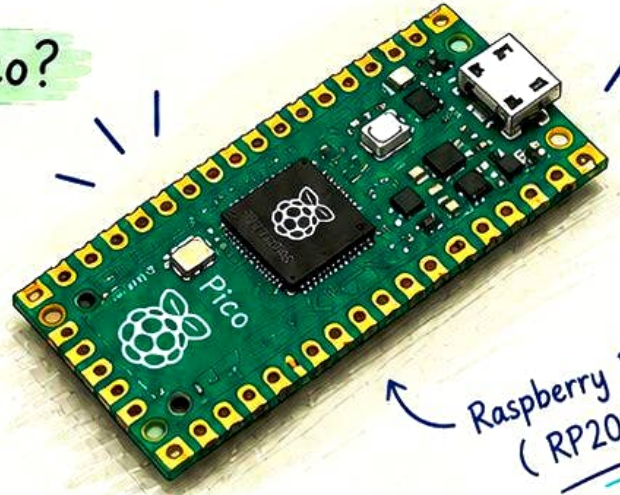
Tiny board...
Big ideas!

Great for
beginners
and
experts!



What is Raspberry Pi Pico?

- Raspberry Pi Pico is a low-cost, high-performance microcontroller board developed by Raspberry Pi. It is based on the RP2040 chip with a dual-core ARM Cortex-M0+ processor.



Raspberry Pi Pico
(RP2040)



Key Features

- Dual-core ARM Cortex-M0+ (up to 133 MHz)
- 264 KB SRAM
- 2 MB onboard flash memory
- 26 multi-function GPIO pins
- Supports MicroPython and C/C++
- Low power consumption



Why Choose Pico?

- High performance at low cost
- Flexible and beginner-friendly
- Rich community support
- Ideal for IoT and embedded projects
- Compact and easy to use



Applications

- IoT devices
- Robotics
- Sensor data logging
- Home automation
- Educational projects

Great
for DIY
projects!



Pinout (Main)

- GPIO Pins : 26 (multi-function)
- ADC Pins : 3 (GP26, 27, 28)
- Flash Memory : 2 MB
- Operating Voltage : 3.3V



"Learn • Build • Create"

Programmable
logic for
real-world
ideas!

Raspberry FPGA

Flexible • Reconfigurable • Powerful

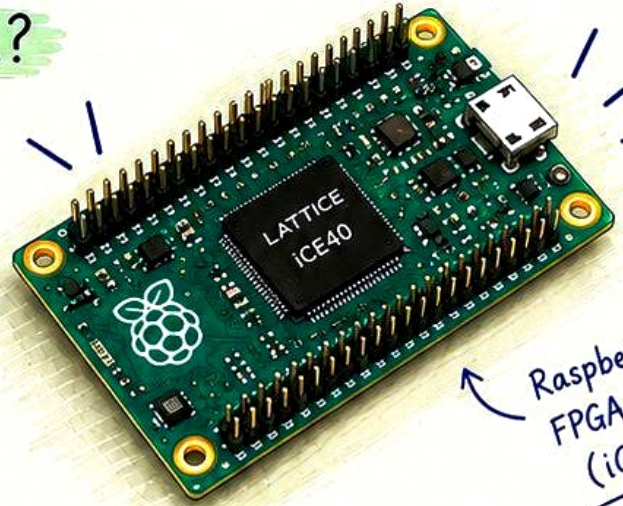
Small board...
Big
possibilities!
😊



What is Raspberry FPGA?

- Raspberry FPGA is a low-cost, high-performance field programmable gate array board developed by Raspberry Pi, based on the **Lattice iCE40** FPGA chip.

It allows users to create and customize digital logic circuits using **Verilog** or **VHDL**.



Raspberry Pi
FPGA Board
(iCE40)



Key Features

- Lattice iCE40 FPGA (up to 3840 logic cells)
- Low power consumption
- Flexible and reconfigurable logic
- On-board GPIO and header pins
- Supports Verilog & VHDL
- Compact and affordable



Why Choose Raspberry FPGA?

- Great for learning digital design and HDL programming
- Ideal for hobbyists, students and embedded projects
- Low cost and easy to use
- Brings your logic ideas to life in hardware



Applications

- Digital logic design & prototyping
- Embedded systems
- IoT devices
- Custom hardware projects
- Education & research

Build
Your own
logic circuits!



Pinout (Main)

- GPIO Pins : 26 (multi-function)
- FPGA I/O : 24 (user I/O)
- Power (3.3V) : 1
- GND : 3

"Learn • Build • Create"

